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F-LOAM: an efficient hybrid model for stock price prediction based on SVM denoising

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capturing temporal features, and the reliance on empirical parameter tuning for hyperparameter optimization, this paper proposes a Feature-enhanced LSTM-Optuna-Augmented Model (F-LOAM). The core innovations of this study are as follows: (1) proposing an enhanced Variational Modal Decomposition (VMD) denoising algorithm, which introduces the Spearman Rank Correlation Coefficient (SRCC) to establish an adaptive modal selection mechanism, thereby improving the signal-to-noise ratio by 23.6%; (2) designing a cascaded Long Short-Term Memory (LSTM)-Light Gradient Boosting Machine (LightGBM) architecture that first extracts high-order feature representations from time series data using LSTM, then utilizes LightGBM's strengths in large-scale data processing for precise predictions, while optimizing both models' hyperparameters through the Optuna framework. In comprehensive evaluations on the self-built MFD12 dataset and public datasets including ACL18 and KDD17, F-LOAM achieves superior performance. Furthermore, the prediction results are converted into executable trading strategies, and backtesting on live markets yield an annualised return of 20.27% and a Sharpe ratio of 1.308, demonstrating

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Stock market data typically exhibit volatility, nonlinearity, non-stationarity, and significant noise [6], posing considerable challenges to accurate forecasting. In financial data analysis, precise signal extraction and noise reduction are essential for enhancing model performance. Arslan et al. [7] developed a hybrid model combining Twitter sentiment analysis with Empirical Mode Decomposition (EMD)-LSTM, demonstrating that integrating social media sentiment indicators with historical price data could substantially improve Bitcoin price prediction accuracy. However, EMD may suffer from mode mixing [8], which can obscure feature extraction. To address this limitation, Zhang et al. [9] proposed Ensemble Empirical Mode Decomposition (EEMD) by introducing white noise to the EMD process, though this approach introduces computational complexity and error accumulation through recursive decomposition. Dragomiretskiy and Zosso [10] subsequently developed Variational Mode Decomposition (VMD) as an effective frequency-domain analysis tool capable of decomposing complex signals into distinct modes with specific center frequencies. Building on these advances, our study proposes an enhanced Spearman-VMD

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learning. In the domain of gradient boosting frameworks, LightGBM has emerged as an efficient solution for large-scale data processing [28]. To optimize model performance, researchers have incorporated metaheuristic algorithms including Particle Swarm Optimization (PSO) [29], Seagull Optimization Algorithm (SOA) [30], and Sparrow Search Algorithm (SSA) [31] for automated hyperparameter tuning. Notably, Wang et al. [32] achieved highly accurate stock price predictions by employing SSA to optimize LSTM hyperparameters. More recently, Optuna has emerged as an automated hyperparameter optimization framework specifically designed to enhance model performance through efficient parameter search [33]. In this study, we utilize Optuna to optimize both LightGBM and LSTM hyperparameters, forming our novel hybrid F-LOAM model.

The principal contributions of this study are threefold:

1. We propose an SVM approach that employs SRCC to filter the IMFs extracted via

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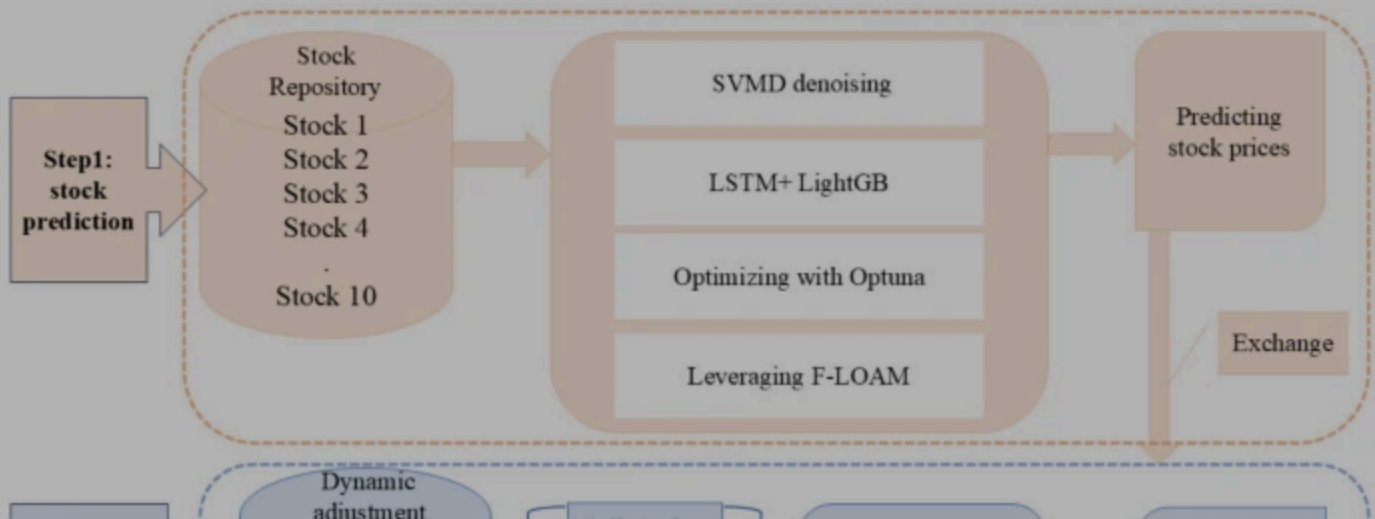
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architecture; Sect. 4 introduces the experimental setup and analyses the results; finally, Sect. 5 summarises the findings and outlines future research directions.

Fig. 1



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simultaneously in both the time domain and frequency domain, their effectiveness highly depends on the selection of wavelet bases and decomposition levels [34], lacking adaptability [35]. Although EMD methods can perform adaptive decomposition based on the data's inherent characteristics, traditional approaches typically discard higher-order IMF components outright, leading to dual issues of effective signal loss and residual noise. These limitations have prompted researchers to explore more refined denoising strategies, such as improved EMD methods combined with threshold processing, to better preserve the effective information of the original data while reducing noise.

In recent years, methods combining multiscale decomposition with correlation analysis have become a research hotspot in financial time series denoising. Empirical studies have shown that improved methods based on EMD can significantly improve the signal-to-noise ratio of stock market volatility signals. Notably, the MIC-EMD method innovatively introduces the maximum information coefficient (MIC) to identify noise-dominated

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machine learning and deep learning techniques, more and more researchers have begun to adopt these advanced models to deal with stock market data. Kumbure et al. [37] reviewed 138 journal articles from 2000 to 2019, Bao et al. [14] reviewed the literature related to data-driven neural networks in the field of stock forecasting from 2015–2023, and this section reviews the unique variables, data processing techniques, various machine learning model variants, and hybrid models used in forecasting.

Niu et al. [38] investigated the impact of geopolitical risk on machine learning models for US stock market volatility prediction using the SHAP algorithm. Their study identified military buildup and war escalation as key factors and demonstrated that the combination of Support Vector Regression (SVR) with trimmed mean achieves superior volatility prediction performance, with a Mean Square Error (MSE) as low as 0.930594. Li et al. [39] proposed a novel DeepAR model (DeepARA) based on the attention mechanism for more accurate and flexible single-point and probabilistic stock price prediction, which utilizes the attention

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Melody-GCN outperforms existing state-of-the-art models on real datasets. Lu and Xu [43] proposed a novel Time-series Recurrent Neural Network (TRNN) model for stock price prediction, which realizes data compression and enhances the price-volume relationship by processing trading volume and time series data to extract trends and inflection points. Zhang et al. [44] proposed a new model called CoATSMP, which adaptively fuses numerical data and textual information to learn time series data in parallel. CoATSMP achieves deep fusion of textual and stock price information through parallel extraction of textual and price features, parameter-level fusion and joint feature processing. Experimental results show that CoATSMP outperforms the baseline model, and its soft fusion method performs better in the framework performs better, and the model containing price and text outperforms the single data source model.

In summary, existing studies have significantly improved the accuracy and robustness of stock price prediction by introducing advanced technologies such as attention mechanisms,

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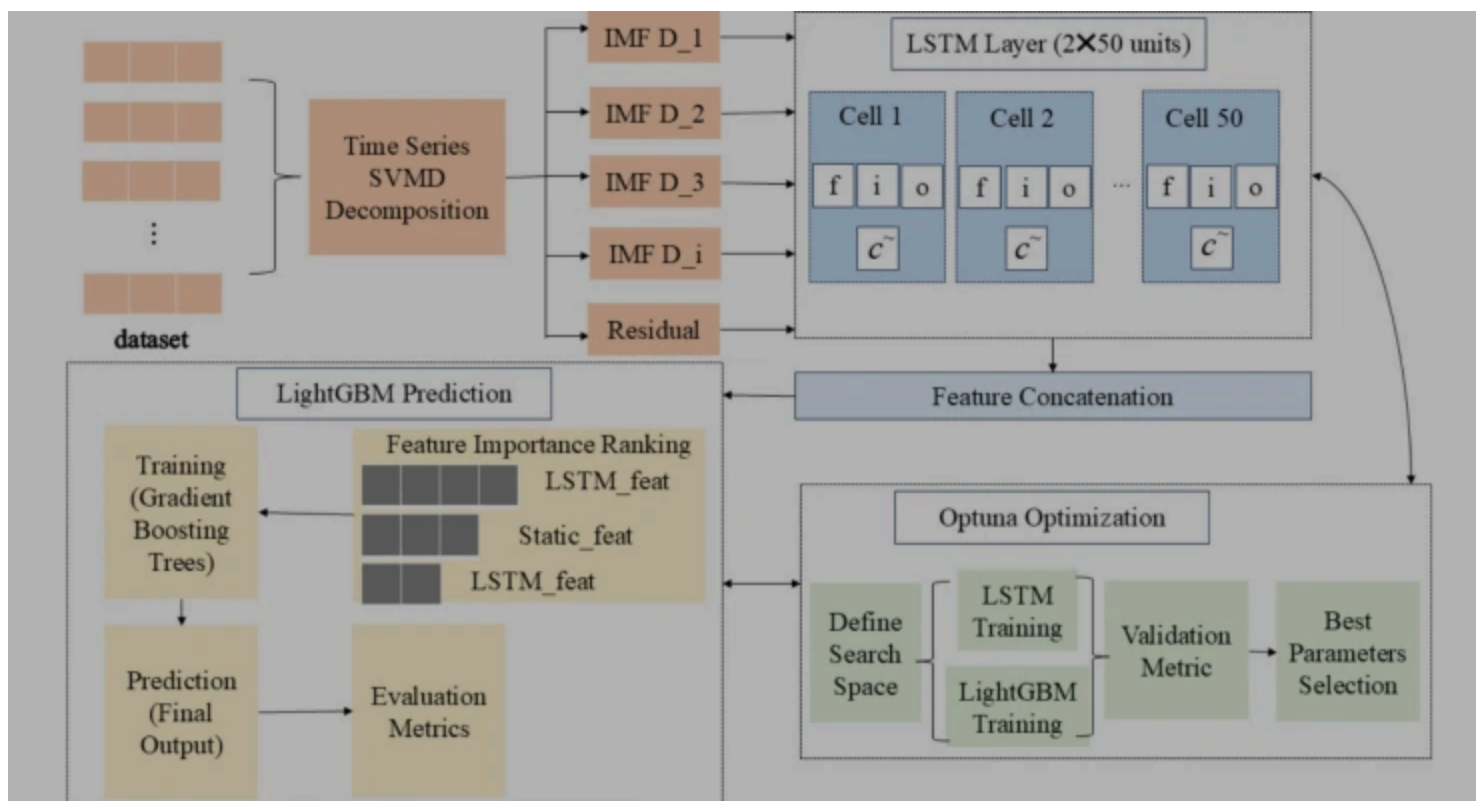
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First, perform signal preprocessing, apply the Hilbert transform to the original signal $(S(t))$ to obtain the analytic signal $(\hat{S}(t))$:

$$\hat{S}(t) = S(t) + jH(S(t)),$$

(1)

where $(H[\cdot])$ denotes the Hilbert transform, and (j) is the imaginary unit.

Second, calculate the single-sided spectrum $(\hat{S}(\omega))$ of $(\hat{S}(t))$:

$$\hat{S}(\omega) = \mathcal{F}\{\hat{S}(t)\},$$

(2)

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Update the IMFs and central frequencies using the Alternating Direction Method of Multipliers (ADMM):

$$\begin{aligned} I_{\{k\}}^{(n+1)} = \arg \min_{\{I_{\{k\}}\}} \bigg[\int \left(\frac{\partial I_{\{k\}}}{\partial t} - \omega_{\{k\}}^{(n)} I_{\{k\}} \right)^2 dt + \frac{\rho}{2} \int \left(I_{\{k\}} - R_{\{k\}}^{(n)} + \lambda_{\{k\}}^{(n)} \right)^2 dt \bigg], \end{aligned}$$

(5)

$$\begin{aligned} \omega_{\{k\}}^{(n+1)} = \arg \min_{\{\omega_{\{k\}}\}} \bigg(\int \bigg(\frac{\partial I_{\{k\}}^{(n+1)}}{\partial t} - \omega_{\{k\}} I_{\{k\}}^{(n+1)} \bigg)^2 dt \bigg), \end{aligned}$$

(6)

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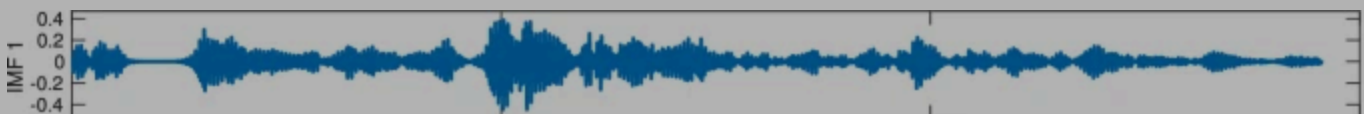
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Reconstruct the signal and form a new dataset $(\hat{\textbf{X}}_{\text{input}} = \begin{Bmatrix} \hat{x}_1, \hat{x}_2, \dots, \hat{x}_8 \end{Bmatrix})$, where $(\hat{x}_1 = S_r(1))$, and so on.

Taking the stock data of 603880.XSHE as an example, the VMD method is first applied to decompose the original data, resulting in seven IMFs, as shown in Fig. 3. Subsequently, the SRCC between each IMF and the original data is calculated, and IMFs with SRCC greater than 0.5 are selected as effective components. Finally, the selected IMFs are reconstructed to generate optimized data with significantly reduced noise [46, 47].

Fig. 3



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3.2 LSTM model

The original concept of LSTM was proposed by Hochreiter in 1991 and further developed by Hochreiter and Schmidhuber in 1997 [48]. Its fundamental element is a scalar memory cell that addresses the vanishing gradient problem by maintaining constant error flow during cell state updates. The memory cell operation is governed by three gating units: the input gate, the forget gate, and the output gate.

To prevent feature bias from occurring during the training process, we standardize the dataset $(\hat{\textbf{X}}_{\text{input}} = \begin{Bmatrix} \hat{x}_1, \hat{x}_2, \dots, \hat{x}_8 \end{Bmatrix})$. The standardization process is outlined below:

First, calculate the mean (μ_j) of each feature (j) in the dataset:

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Normalize the data $(\hat{X}_{\text{input}} = \{\hat{x}_1, \hat{x}_2, \dots, \hat{x}_8\})$ and reshape it into a form suitable for LSTM model input. In the LSTM model, the memory cell at time step (t) is updated following specific rules:

$$\begin{aligned} c_t = f_t c_{t-1} + i_t z_t, \end{aligned}$$

(12a)

$$\begin{aligned} h_t = o_t \tilde{h}_t, \quad \tilde{h}_t = \psi(c_t), \end{aligned}$$

(12b)

$$\begin{aligned} z_t = \varphi \left(\text{w}_z^{\text{top}} \hat{X}_{\text{input}} + r_z h_{t-1} + b_z \right), \end{aligned}$$

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h_{t-1}) to the cell input, input gate, forget gate, and output gate. b_z, b_i, b_f, b_o are the corresponding bias vectors, σ is the sigmoid activation function, and ψ is the $\tanh$ function.

After extracting features from the eight input variables using LSTM, a fixed-length feature vector encapsulating temporal information and data patterns is obtained. This feature vector serves as input to downstream models for subsequent analysis and prediction. It consists of two LSTM layers, each containing 50 memory units.

The LSTM component effectively captures complex patterns and long-term dependencies in time series data, while LightGBM efficiently processes the extracted features and generates rapid predictions. In this study, the output from the LSTM is fed into LightGBM to enhance overall model performance by leveraging its computational efficiency. The characteristics and applications of LightGBM are detailed in Sect. [3.4](#).

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where $(z_1, z_2, \dots, z_8)$ are the normalized features, and (h_t) is the hidden state of the LSTM. Using the advantages of LightGBM in processing high-frequency data, predictions are made with the feature vectors.

LightGBM uses a Classification And Regression Tree (CART) regression tree as the base learner, and the final model divided by the overall (m) -tree can be expressed as:

$$\hat{y}_i = \sum_{t=1}^m f_t(d_i), \quad (14)$$

where (f_t) is the first (t) base learner and $(\hat{y}_i)$ is the prediction result of the first (i) sample. The objective function of LightGBM algorithm is as follows:

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In order to optimize the objective function, LightGBM expands the objective function to the second order and the expanded function can be expressed as:

$$\begin{aligned} \text{Obj}^{\{t\}} = & \sum_{i=1}^n \left(l \left(y_i, \hat{y}_i^{\{t-1\}} \right) + g_i f_t(d_i) + \frac{1}{2} h_i f_t^2(d_i) \right) + \sum_{i=1}^t \Omega(f_i) + C, \end{aligned}$$

(18)

where (C) is the constant term, and (g_i) and (h_i) denote the first-order and second-order gradient statistics of the loss function, respectively, which can be expressed as:

$$\begin{aligned} g_i = & \frac{\partial l \left(y_i, \hat{y}_i^{\{t-1\}} \right)}{\partial \hat{y}_i^{\{t-1\}}}, \end{aligned}$$

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$\mid q(d_i)=j \setminus)$ is the sample set belonging to the first $\setminus (j \setminus)$ leaf node, the objective function is:

$$\begin{aligned} \text{Obj}^{\{t\}} = \sum_{j=1}^T \left(\left(\sum_{i \in I_j} g_i \right) w_j + \frac{1}{2} \left(\sum_{i \in I_j} h_i + \right) w_j^2 \right) + \gamma T. \end{aligned}$$

(22)

To simplify the objective function, we define $\setminus (G_j = \sum_{i \in I_j} g_i \setminus)$, $\setminus (H_j = \sum_{i \in I_j} h_i \setminus)$, and the objective function is:

$$\begin{aligned} \text{Obj}^{\{t\}} = \sum_{j=1}^T \biggl(G_j w_j + \frac{1}{2} (H_j + \lambda) w_j^2 \biggr) + \gamma T. \end{aligned}$$

(23)

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\end{aligned}\$\$

(26)

Let L and R denote the sample sets of the left and right branches, respectively. To minimize the loss function, LightGBM employs a leaf-wise growth strategy that splits the leaf nodes with the largest gain. Although the leaf-by-leaf strategy generates a small number of leaf nodes, it may produce a deep decision tree and lead to overfitting. In addition, hyperparameters such as the number of base learners and the values of γ and λ have an important impact on the classification performance of the LightGBM algorithm. In order to obtain better performance of the algorithm, it is necessary to choose appropriate hyperparameters for LightGBM as described in Sect. 4.1.

3.4 Automatic optimization of hyperparameters using the

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Step 2: Define the search space for hyperparameters, specifying the type and possible range of values for each hyperparameter.

For the LightGBM model, the hyperparameters include:

$$\begin{aligned} & \left\{ \begin{array}{l} H_{\text{num_leaves}} \in \{20, 30, 40, 50\}, \\ H_{\text{learning_rate}} \in [0.01, 0.1], \\ H_{\text{feature_fraction}} \in [0.8, 1.0], \\ H_{\text{bagging_fraction}} \in [0.8, 1.0], \\ H_{\text{bagging_freq}} \in \{1, 5, 10\}. \end{array} \right. \end{aligned}$$

(28)

For the LSTM model, the hyperparameters include:

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4 Experimental setup and benchmarking strategy on test dataset

4.1 Experimental set-up

This section consists of two parts: first, a description of the dataset used and its sources; second, a detailed description of the data preprocessing process using SVDM. The experiments were conducted in a local hardware environment equipped with an AMD Ryzen 7 8745 H processor, Hynix DDR5 5600MHz memory, and an NVIDIA GeForce RTX 4060 Laptop GPU, which ensured efficient computational performance. F-LOAM was developed in PyCharm using Python 3.12.5, and the implementation with TensorFlow and Keras APIs was completed. Subsequently, simulated trading and backtesting were performed in the JoinQuant platform based on the model predictions. F-LOAM specific parameter configurations can be found in Table [1](#).

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To comprehensively evaluate model performance, we adopted a two-stage experimental design: first, model training and testing were conducted on the self-constructed MFD12 dataset, followed by validation of the model's generalization ability using the publicly available datasets ACL18 [51] and KDD17 [52]. The ACL18 dataset covers 88 cross-industry stocks traded on the NASDAQ and NYSE from 2014 to 2016 (with a ratio of 51.73% and 48.27% for rising and falling samples, respectively), while the KDD17 dataset includes historical data for 50 stocks in the US stock market from 2007 to 2016.

In terms of experimental data processing, we used a random partitioning method to divide each original dataset into a training set and a test set, with the test set accounting for 20% and the remaining 80% of the data automatically allocated to the training set.

4.1.2 Data pre-processing

This paper uses three stocks from the MFD12 dataset (603880.XSHE, 002830.XSHG, and

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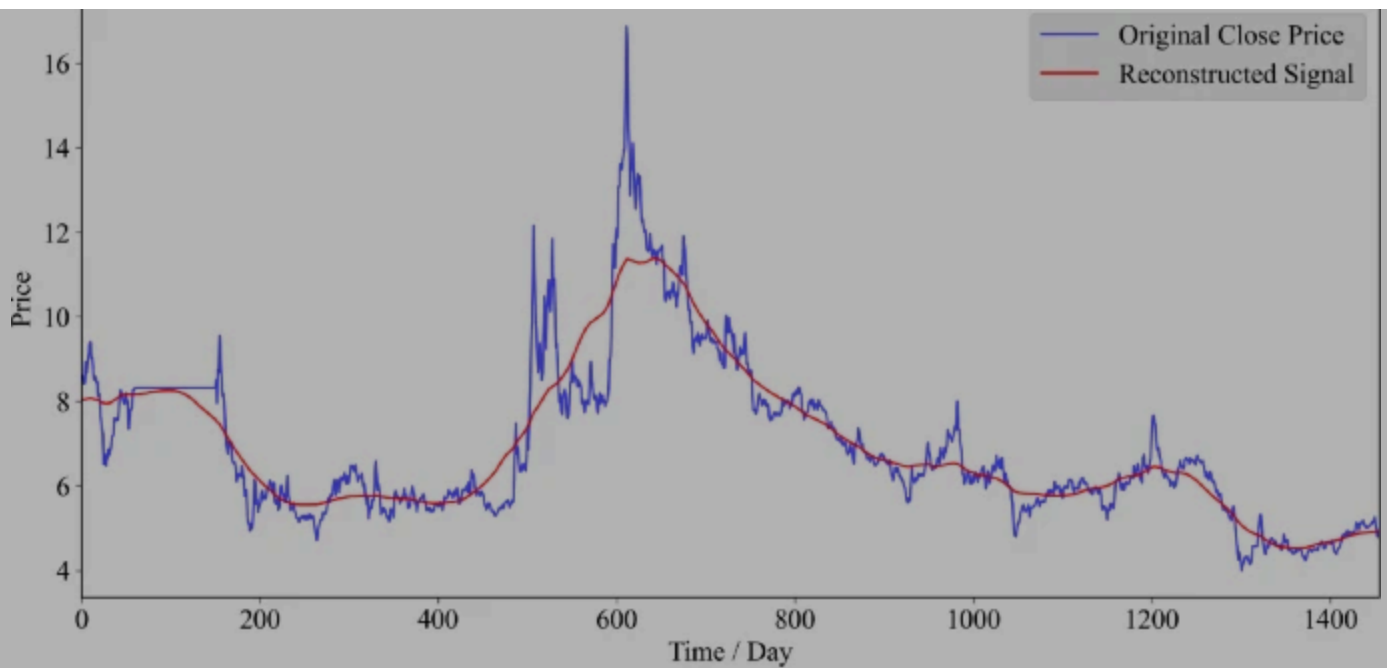
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(a)

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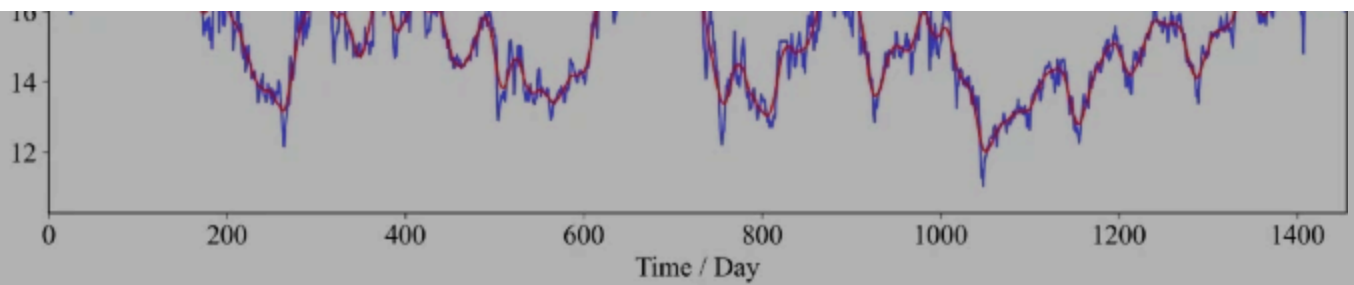
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(c)

The experiment uses SVMd to denoise financial time series data. with a selection of denoised data presented for illustration. **a** The original closing price data and denoised data for 603880.XSHE; **b** the closing price sequence for 002830.XSHG; **c** the closing price sequence for 300717.XSHG

4.2 Experimental design of F-LOAM based on SVMd noise reduction

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$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|,$$

(30)

$$R^2 = \frac{1 - \sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}.$$

(31)

To validate the superiority of the model's performance, we conducted comparative experiments between F-LOAM and the hybrid models SG-SSA-LSTM and CNN-LSTM. The detailed results on the MFD12 dataset are shown in Table 3, while the comparative results on the ACL18 and KDD17 datasets are shown in Fig. 5.

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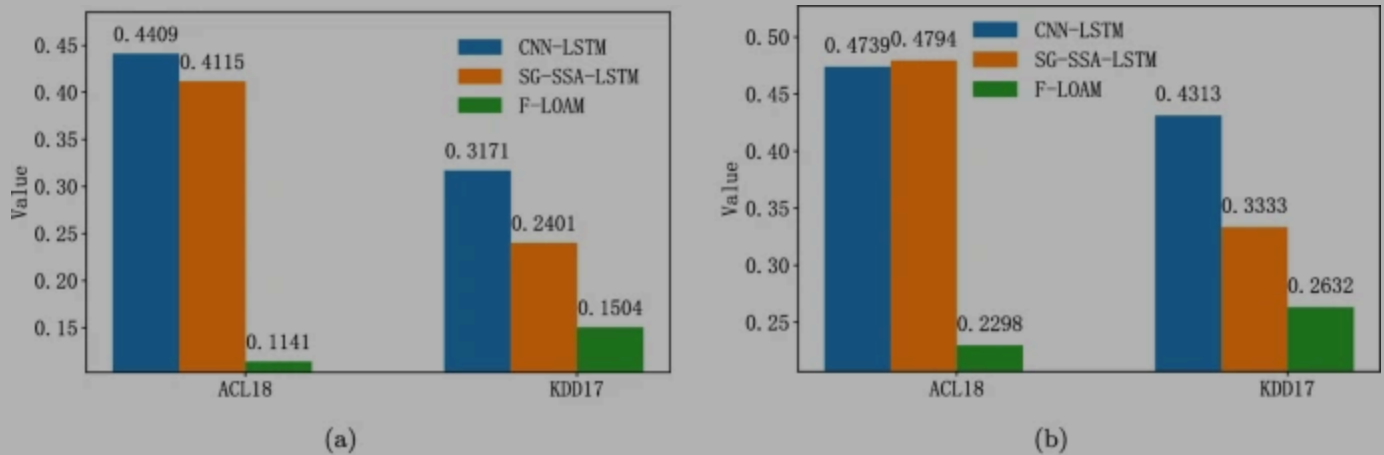
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simultaneously capture long-term trends and short-term volatility characteristics in financial time series.

Fig. 5



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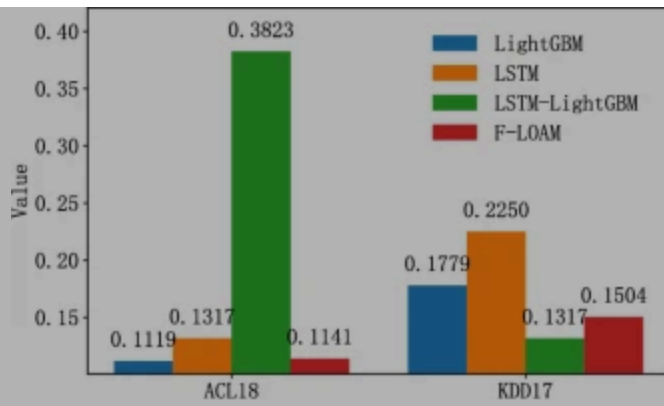
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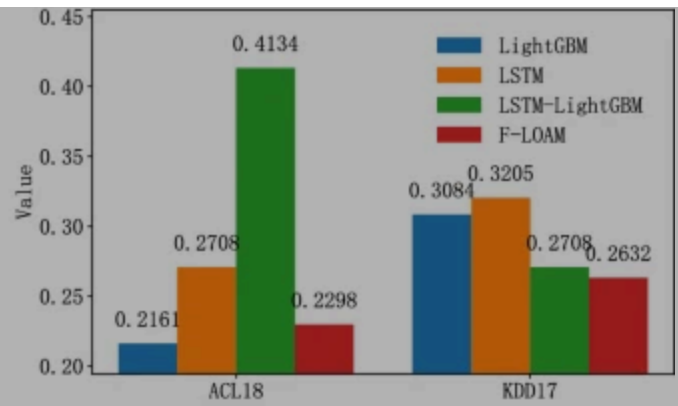
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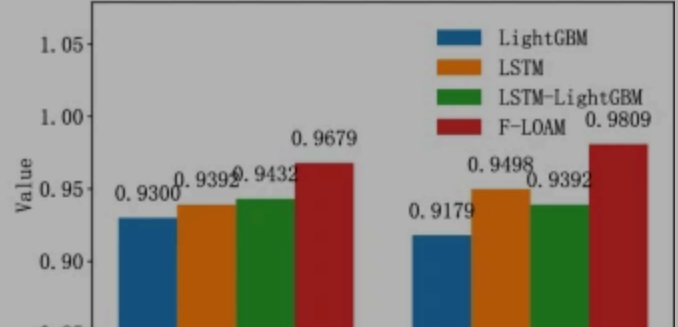
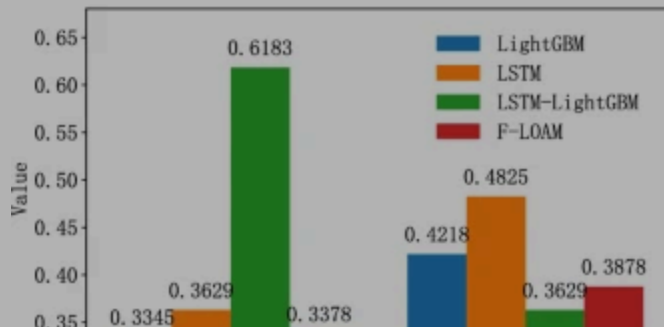
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(a)



(b)



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Cumulative returns on dynamic position adjustment strategies using model forecasts and without model forecasts between 2024-06 and 2024-09

4.2.3 Ablation experiments

To systematically evaluate the contributions of each module in the F-LOAM model, on the ACL18 and KDD17 datasets, we conducted a comprehensive ablation experiment study. As shown in Fig. 6, the base LightGBM model outperforms LSTM in feature learning. The LSTM-LightGBM ensemble model achieves a 26.4% performance improvement, validating the complementary nature of temporal features and gradient boosting. The introduction of Optuna hyperparameter optimisation further enhanced the complete F-LOAM model's performance by 22.1%. It maintained optimal results across all cross-dataset evaluations, achieving a minimum MSE of 0.064131 and a maximum RMSE of 0.980871.

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conventional models across multiple datasets. Empirical validation through trading strategy backtesting achieved a 20.27% cumulative return and 1.308 Sharpe ratio, significantly outperforming benchmark strategies and confirming its practical value for intelligent investment research.

Data availability

No datasets were generated or analysed during the current study.

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Contributions

L.L: Conceptualization, Preparation, Funding acquisition, Supervision, Writing–review and editing. J. L: Data curation, Formal analysis, Methodology, Investigation, Visualization, Writing–original draft.

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